

● Strong and Weak ties Clustering coefficient

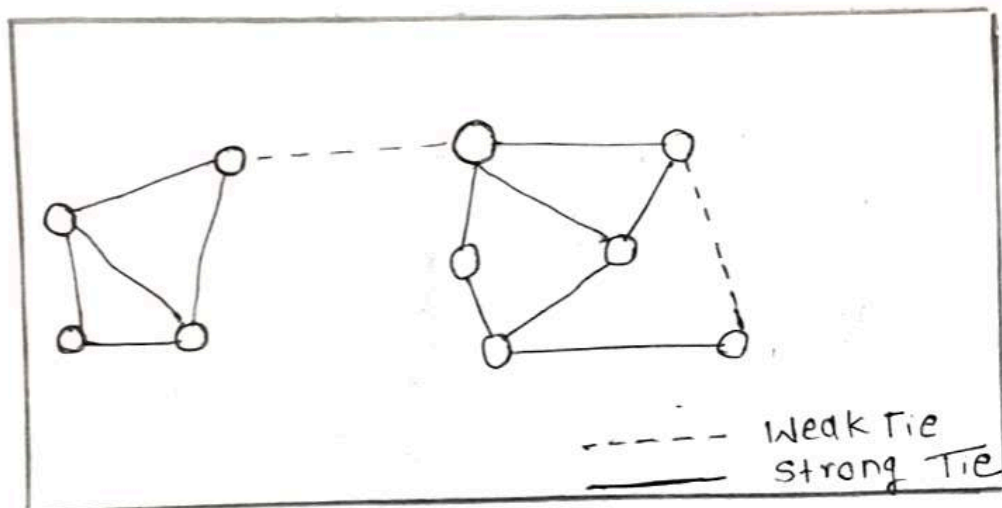
Strong relationship exists betⁿ close members with frequent interactions or meetings.

Ex. Strong ties :-

family members and close friends cause strong ties.

Weak Tie :-

Weak relationship is caused by distant social relationships and very frequent meetings or interactions.



strong ties :-

(i.e., close relationships with friends and family)

Weak ties :-

(i.e., more distant or casual relationships)

Q. What is the difference between strong and weak relationship in social networking?

Q. What is the role of weak ties?
(Your weak ties connect you to networks that are outside of your own circle.)

Q. Write a note on strong and weak ties.

- ① Weak ties are people you know but not very well. They are merely acquaintances. You interact with these people at work and in your personal life only as much as you need to.
- ② strong ties are folks you know very well

③ Advantages of strong ties :-

- 1) less easily broken and pass a higher volume of information than weak ties.
- 2) strong ties are thought to be twice as effective at passing information betⁿ two people.

Disadvantages of strong ties :-

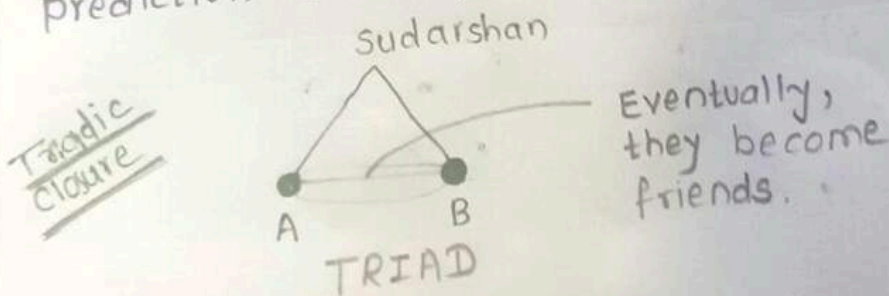
- 1) The disadvantage of strong ties at this stage is that we might receive more mediocre ideas rather than more stimulating and varied perspectives from weak ties. In the phase of idea elaboration, strong ties would become more reliable.

clustering coefficient :-

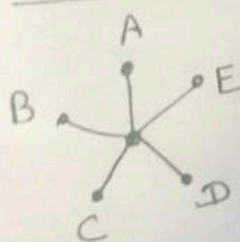
- ① The clustering coefficient measures how connected a vertex's neighbours are to one another. More specifically, it is calculated as :

$$\frac{\text{No. of edges connecting a vertex's neighbors}}{\text{the total no. of possible edges bet. the vertex's neighbors.}}$$

- ② clustering coefficient is a property of a node in a network. Roughly speaking it tells how well connected the neighborhood of the node is. if the neighborhood is fully connected, the clustering coefficient is 1 and (a value close to 0 means that there are hardly any connections in the neighborhood.)
- ③ A clustering coefficient is a measure of degree to which nodes in a graph tend to cluster together. clustering coefficient of a network plays vital role to influence the behavior of the link prediction technique.



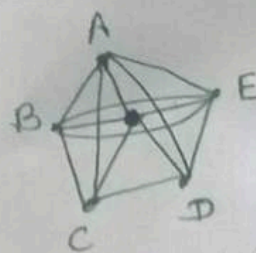
Scenario - 1



clustering coefficient = 1

int

Scenario - 2



clustering coefficient = 0

Total possible no. of friendships

$$= \frac{5 \times 4}{2} = \frac{n(n-1)}{2}$$

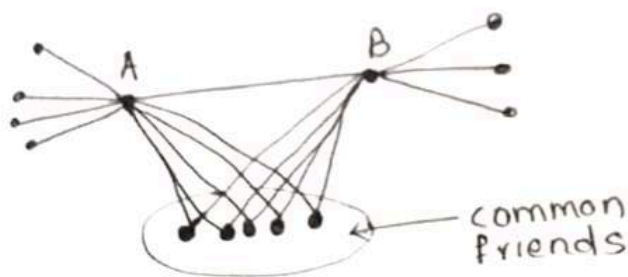
$$= \frac{20}{2}$$

$$= 10$$

$\frac{6}{10} \Rightarrow$ clustering Coefficient

$$\frac{N}{D} = \frac{\text{Total no of friendship bet}^n \text{ your frd.}}{\text{Total possible friendship}}$$

① Neighborhood overlap :-

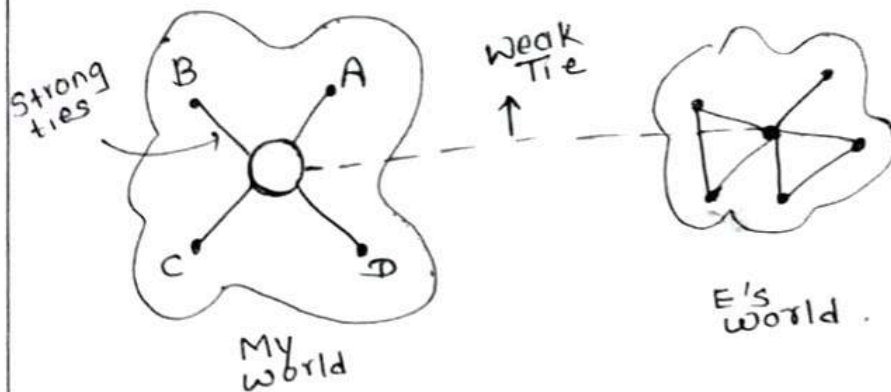


$$= \frac{\text{Common friends}}{\text{Total friends}}$$

⇒ closed to zero → Neighborhood overlap is min
 closed to One → Neighborhood overlap is max

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Q. illustrate Granovetter's strength of weak ties.



Assume I have 5 friends A, B, C, D & E. out of these 5 friends 4 (A, B, C, D) work in the same place where I work and E works in a different company. Now almost all the information that A, B, C, & D have is already known to me as we live in the same world. But E is from a different world and that is why every information that E has is new

to me although E has a weak tie with me
 So E is the one who helps me in getting new
 information about the different world which
 increases my knowledge.

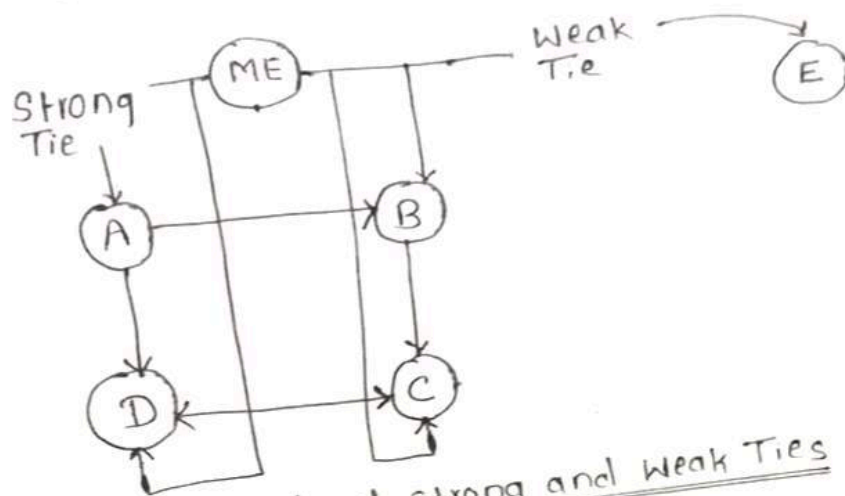


Fig. Example of strong and weak ties

so strong ties are weak when it comes to
 information related to new jobs or job switch
 and weak ties are strong in the same scenario.

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Q Explain Community Detection Using Girvan Newman Algorithm.

- The Girvan-Newman algorithm detects communities by progressively removing edges from the original network.
- The connected components of the remaining network are the communities.
- The algorithm's steps for community detection are summarized below.

1. The betweenness of all existing edges in the network is calculated first.
2. The edge(s) with the highest betweenness are removed.
3. The betweenness of all edges affected by the removal is recalculated.
4. Steps 2 and 3 are repeated until no edges remain.

- First step of the Girvan-Newman Algorithm:-

We shall describe a method called the Girvan-Newman (GN) Algorithm, which visits each node x once and computes the no. of shortest paths from x to each of the other nodes that go through each of the edges. the algorithm begins by performing a breadth first search (BFS) of the graph, starting at the node x .

- Girvan-Newman community structure :-

The Girvan-Newman method for the detection and analysis of community structure is based on the iterative elimination of edges with the highest no. of the shortest paths that go through them.

By eliminating edges the networks breaks down into smaller networks i.e., communities.

- Girvan Newman method for community detection in social Network :-

- The Girvan Newman algorithm detects communities by progressively removing edges from the original n/w.
- the connected components of the remaining n/w are the communities.

— x —

Girvan-Newman algorithm

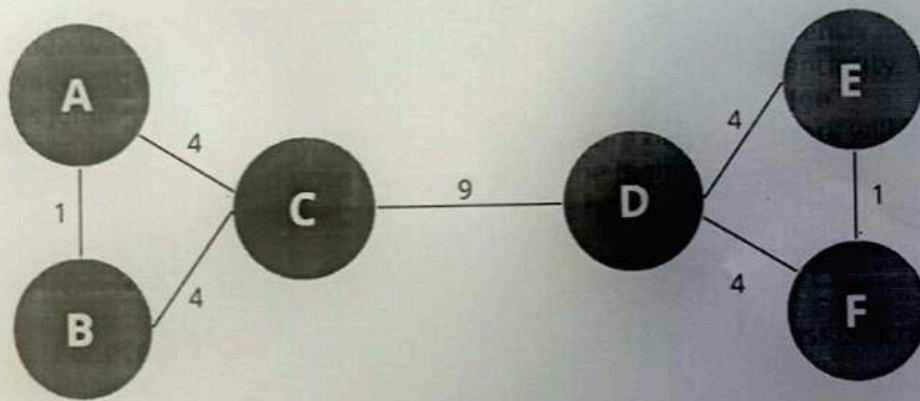
The **Girvan-Newman algorithm** for the detection and analysis of community structure relies on the iterative elimination of edges that have the highest number of shortest paths between nodes passing through them. By removing edges from the graph one-by-one, the network breaks down into smaller pieces, so-called communities. The algorithm was introduced by Michelle Girvan and Mark Newman.

How does it work? __

The idea was to find which edges in a network occur most frequently between other pairs of nodes by finding edges betweenness centrality. The edges joining communities are then expected to have a high edge betweenness. The underlying community structure of the network will be much more fine-grained once the edges with the highest betweenness are eliminated which means that communities will be much easier to spot.

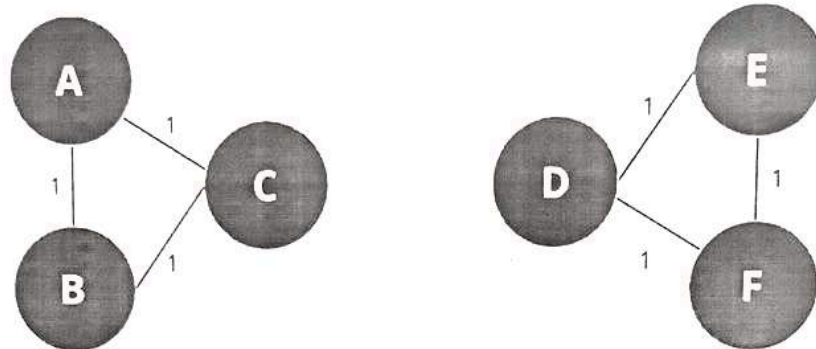
The Girvan-Newman algorithm can be divided into four main steps:

1. For every edge in a graph, calculate the edge betweenness centrality.
2. Remove the edge with the highest betweenness centrality.
3. Calculate the betweenness centrality for every remaining edge.
4. Repeat steps 2-4 until there are no more edges left.



In this example, you can see how a typical graph looks like when edges are assigned weights based on the number of shortest paths passing through them. To keep things simple, we only calculated the number of undirected shortest paths that pass through an edge. The edge between

nodes **A** and **B** has a strength of 1 because we don't count **A**->**B** and **B**->**A** as two different paths.

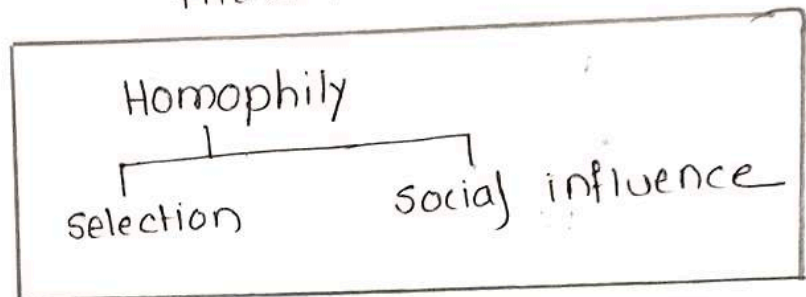


The Girvan-Newman algorithm would remove the edge between nodes **C** and **D** because it is the one with the highest strength. As you can see intuitively, this means that the edge is located between communities. After removing an edge, the betweenness centrality has to be recalculated for every remaining edge. In this example, we have come to the point where every edge has the same betweenness centrality.

③ Selection and social influence :-

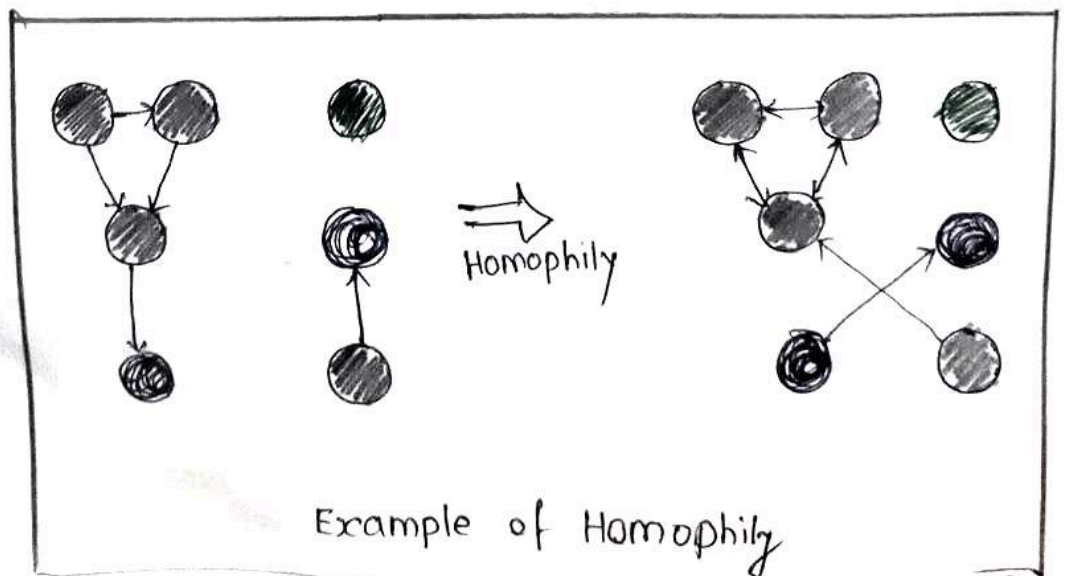
selection :- the tendency of people to form friendships with others with similar characteristics.

social influence :- people may modify their behaviors to bring them more closely into alignment with the behaviors of their friends.



③ Homophily :-

- ① Homophily is the tendency in social groups of similar people connected together.
- ② We often hear similar voices interact with like-minded people.
- ③ Homophily has a significant impact on social media.
- ④ Ex. - Birds with feather flock together.



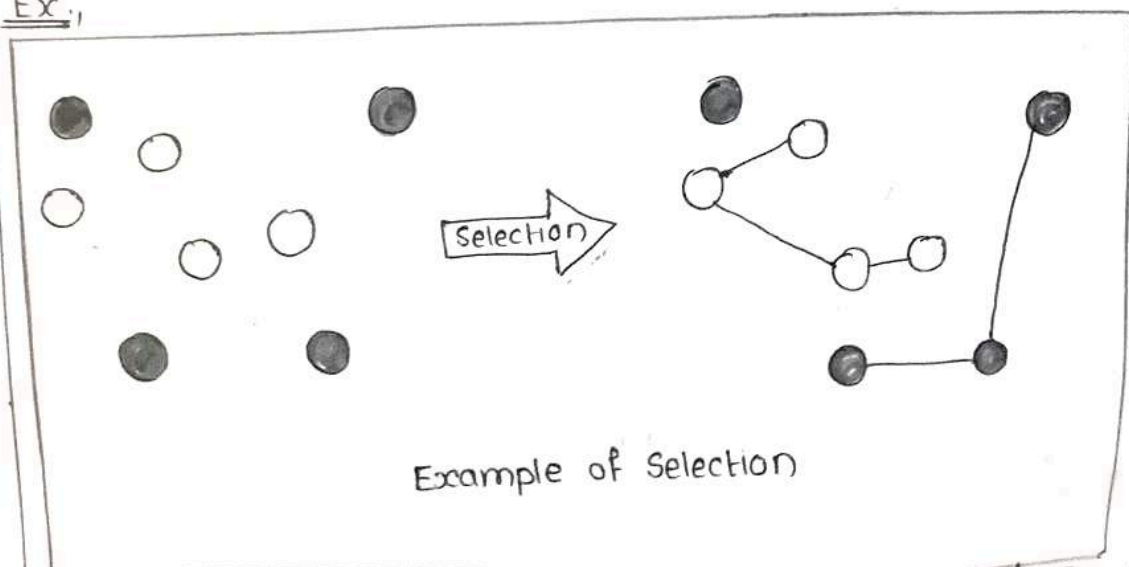
- Assume there are 1000 people in a party out of which 500 are of age ranges from 18-25 and other 500 people are from the age group 40-50. So mathematically if we pick any friendship randomly then mostly it will be one teenager and one middle-aged person which is most probable condition.
- But we all know that teenagers may want to talk to a teenager and middle-aged may want to talk to middle-aged people. This is known as Homophily.

- Ex
- 1000 people
 - 500 (18-25) teenager
 - 500 (40-50) middle-aged

● Selection :-

- ① It is the tendency in which people make friends with similar interests.
i.e., people select other people having similar habits or interests.
- ② In selection, people select similar nodes and connects with them.

Ex.,



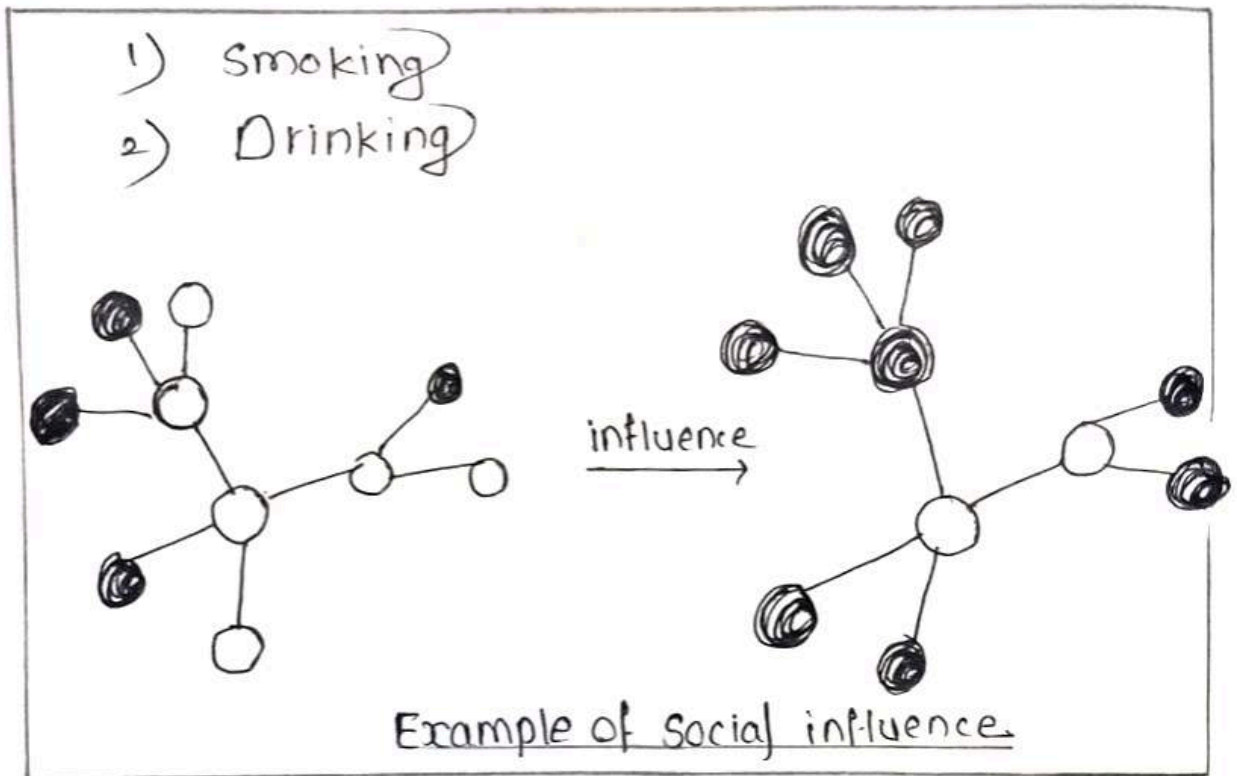
Example of Selection

Suppose there is a person who speaks Spanish and I also know Spanish so I will select that person and talk to him. This is called selection

⑩ Social Influence :-

- It is the tendency in which people change their attitude or behavior to meet the social environment by getting influenced by other people is called social influence.
- social influence makes connected nodes similar

Ex.



Suppose I have a friend who smokes so he will influence me to smoke. This is social influence.

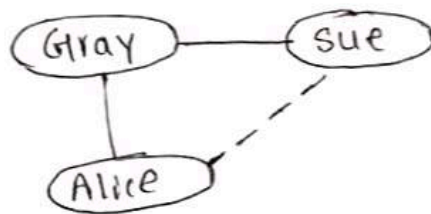
their

Foci Closure and Membership Closure :-

Different types of closures in social networks

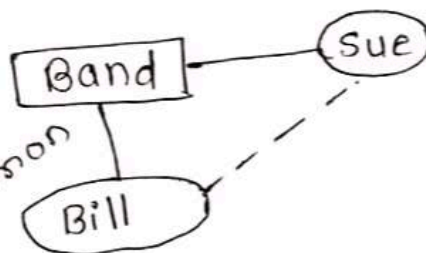
- 1) Triadic closures
- 2) Focal closures
- 3) Membership closure.

1) Triadic closures :-

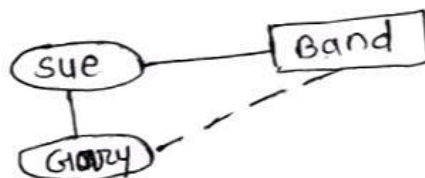


Due to a common
frd or
homophily
↑
Triadic closure

Focal closure
new
edge may
form
because
of a common
focal
point.



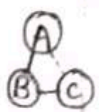
Focal closure :
closure due to
selection.



Membership closure :
closure due to
social influence.

Closure in Networking) :-

A social network perspective that illustrates. network benefits from strongly interconnected social ties in the closed network.



Ex, of Triadic closures

if there were three people within a network, A, B, & C, if A & B are friends, and B & C are friends there is a high possibility that A & C are also frd. if each person were represented as a node, these 3 nodes combined would form a triadic closure.

foci of an ellipse :-

- 1) the foci of an ellipse are two fixed, interior points.
- 2) Foci are the two points on the ellipse. The
- 3) The distance around the ellipse is called the perimeter.
- 4) The area of an ellipse can be defined as the total no. of square units that it takes to fill up the region inside an ellipse.

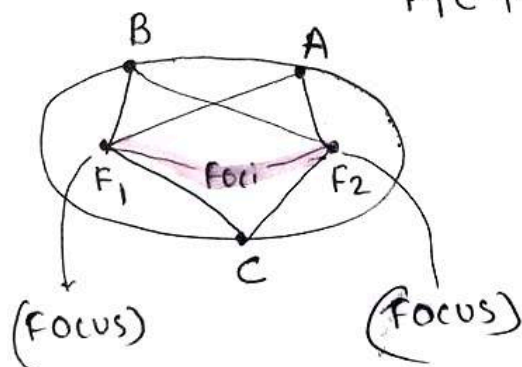
F_1 F_2 $\Rightarrow$ F_1 & F_2
two points

set of points $\Rightarrow$ Sum of distances
from F_1 & F_2
is Constant (α)

$$F_1A + F_2A = \alpha$$

$$F_1B + F_2B = \alpha$$

$$F_1C + F_2C = \alpha$$



$\Rightarrow$ Collection of all points
in a plane such that
the sum of distances
of each point from
two fixed points in
the plane is constant.

* foci closure $\rightarrow$

it is the probability of becoming
friends when they have same foci.

IMP

Schelling's model of Segregation :-

→ Check Google classroom

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